where geodesics exist, obviously. It is in this case a continuous (locally Lipschitz) curve, which coincides with the piecewise constant interpolation x^τ at times $t = k\tau$.

De Giorgi, in [138], defined[1] the notion of *Minimizing Movements* (see also [7]):

Definition 8.3. A curve $x : [0,T] \to X$ is said to be a Minimizing Movement if there exists a sequence of time steps $\tau_j \to 0$ such that the the piecewise constant interpolations x^{τ_j}, built from a sequence of solutions of the iterated minimization scheme (8.1), uniformly converge to x on $[0,T]$.

Compactness results guaranteeing the existence of Minimizing Movements are derived from a simple property giving a Hölder behaviour for the curves x^τ: for every τ and every k, the optimality of x^τ_{k+1} provides

$$F(x^\tau_{k+1}) + \frac{d(x^\tau_{k+1}, x^\tau_k)^2}{2\tau} \le F(x^\tau_k), \tag{8.2}$$

which implies

$$d(x^\tau_{k+1}, x^\tau_k)^2 \le 2\tau \left(F(x^\tau_k) - F(x^\tau_{k+1})\right).$$

If $F(x_0)$ is finite and F is bounded from below, taking the sum over k, we have

$$\sum_{k=0}^{l} d(x^\tau_{k+1}, x^\tau_k)^2 \le 2\tau \left(F(x^\tau_0) - F(x^\tau_{l+1})\right) \le C\tau.$$

Given $t \in [0,T]$, denote by $\ell(t)$ the unique integer such that $t \in](\ell(t)-1)\tau, \ell(t)\tau]$. For $t < s$ apply the Cauchy-Schwartz inequality to the sum between the indices $\ell(t)$ and $\ell(s)$ (note that we have $|\ell(t) - \ell(s)| \le \frac{|t-s|}{\tau} + 1$)

$$\begin{aligned} d(x^\tau(t), x^\tau(s)) &\le \sum_{k=\ell(t)}^{\ell(s)-1} d(x^\tau_{k+1}, x^\tau_k) \le \left(\sum_{k=\ell(t)}^{\ell(s)-1} d(x^\tau_{k+1}, x^\tau_k)^2\right)^{\frac{1}{2}} |\ell(s) - \ell(t)|^{\frac{1}{2}} \\ &\le C\sqrt{\tau} \frac{(|t-s| + \tau)^{\frac{1}{2}}}{\sqrt{\tau}} \le C\left(|t-s|^{1/2} + \sqrt{\tau}\right). \end{aligned}$$

This means that the curves x^τ - if we forget for a while that they are actually discontinuous - are morally equi-Hölder continuous of exponent $1/2$. The situation is even clearer for $\widetilde{x}^\tau$. Indeed, we have

$$\tau \left(\frac{d(x^\tau_{k-1}, x^\tau_k)}{\tau}\right)^2 = \int_{(k-1)\tau}^{k\tau} |(\widetilde{x}^\tau)'|^2(t)\mathrm{d}t,$$

which implies, by summing up

$$\int_0^T |(\widetilde{x}^\tau)'|^2(t)\mathrm{d}t \le C.$$

[1] We avoid here the distinction between Minimizing Movements and Generalized Minimizing Movements, which is not crucial in our analysis.

This means that the curves $\widehat{x^\tau}$ are bounded in $H^1([0,T];X)$ (the space of absolutely continuous curves with square-integrable metric derivative), which also implies a $C^{0,\frac{1}{2}}$ bound by usual Sobolev embedding.

If the space X, the distance d, and the functional F are explicitly known, in some cases it is already possible to pass to the limit in the optimality conditions of each optimization problem in the time-discrete setting, and to characterize the limit curve $x(t)$. It will be possible to do so in the space of probability measures, which is the topic of this chapter, but not in many other cases. Indeed, without a little bit of (differential) structure on X, it is almost impossible. If we want to develop a general theory of gradient flows in metric spaces, we need to exploit finer tools, able to characterize, with the only help of metric quantities, the fact that a curve $x(t)$ is a gradient flow.

We present here an inequality which is satisfied by gradient flows in the smooth Euclidean case, and which can be used as a definition of gradient flow in the metric case, since all the quantities which are involved have their metric counterpart. Another inequality, called EVI, will be presented in the discussion section (8.4.1).

The first observation is the following: thanks to the Cauchy-Schwartz inequality, for every curve $x(t)$ we have

$$\begin{aligned} F(x(s))-F(x(t)) = \int_s^t -\nabla F(x(r))\cdot x'(r)\,dr &\le \int_s^t |\nabla F(x(r))||x'(r)|\,dr \\ &\le \int_s^t \left(\frac{1}{2}|x'(r)|^2+\frac{1}{2}|\nabla F(x(r))|^2\right)dr. \end{aligned}$$

Here, the first inequality is an equality if and only if $x'(r)$ and $\nabla F(x(r))$ are vectors with opposite directions for a.e. r, and the second is an equality if and only if their norms are equal. Hence, the condition, called EDE (*Energy Dissipation Equality*)

$$F(x(s))-F(x(t)) = \int_s^t \left(\frac{1}{2}|x'(r)|^2+\frac{1}{2}|\nabla F(x(r))|^2\right)dr, \quad \text{for all } s<t$$

(or even the simple inequality $F(x(s))-F(x(t)) \ge \int_s^t \left(\frac{1}{2}|x'(r)|^2+\frac{1}{2}|\nabla F(x(r))|^2\right)dr$) is equivalent to $x' = -\nabla F(x)$ a.e., and could be taken as a definition of gradient flow. This is what is done in a series of works by Ambrosio, Gigli and Savaré, and in particular in their book [17]. Developing this (huge) theory is not among the scopes of this book. The reader who is curious about this abstract approach but wants to find a synthesis of their work from the point of view of the author can have a look at [275]. The role of the different parts of the theory with respect to the possible applications is clarified as far as possible[2] (we also discuss in short some of these issues in Section 8.4.1).

8.2 Gradient flows in $\mathbb{W}_2$, derivation of the PDE

In this section we give a short and sketchy presentation of what can be done when we consider the gradient flow of a functional $F:\mathscr{P}(\Omega)\to\mathbb{R}\cup\{+\infty\}$. The functional will

[2] Unfortunately, some knowledge of French is required (even if not forbidden, English is unusual in the Bourbaki seminar, since “Nicolas Bourbaki a une préférence pour le français”).

be supposed l.s.c. for the weak convergence of probabilities and Ω compact (which implies that F admits a minimizer, and is bounded from below). In particular, we will give an heuristic on how to derive a PDE from the optimality conditions at each time step. This is obviously something that we can only do in this particular metric space and does not work in the general case of an abstract metric space.

Indeed, we exploit the fact that we know, from Chapter 5, that all absolutely continuous curves in the space $\mathbb{W}_2(\Omega)$ are solution of the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0$, for a suitable vector field $\mathbf{v}$. The goal is to identify the vector field $\mathbf{v}_t$, which will depend on the measure ρ_t, in a way which is ruled by the functional F.

We consider the iterated minimization scheme

$$\rho^\tau_{(k+1)} \in \operatorname{argmin}_\rho \; F(\rho) + \frac{W_2^2(\rho, \rho^\tau_{(k)})}{2\tau}. \tag{8.3}$$

Each of these problems has a solution, by compactness of $\mathscr{P}(\Omega)$ and semi-continuity of the objective function. Exactly as we did for the Euclidean case (F defined on $\mathbb{R}^d$), we want to study the optimality conditions of these problems so as to guess the limit equation.

We recall the notation $\frac{\delta G}{\delta \rho}(\rho)$ for the first variation of a functional $G : \mathscr{P}(\Omega) \to \mathbb{R} \cup \{+\infty\}$. We will suppose that $\frac{\delta F}{\delta \rho}$ is known and try to write the limit equation in terms of this operator.

Now, take an optimal measure $\widehat{\rho}$ for the minimization problem at step k and compute variations with respect to perturbations of the form $\rho_\varepsilon := (1-\varepsilon)\widehat{\rho} + \varepsilon \widetilde{\rho}$, where $\widetilde{\rho}$ is any other probability measure. Using Proposition 7.20 we obtain

$$\frac{\delta F}{\delta \rho}(\rho) + \frac{\varphi}{\tau} = constant,$$

whre φ is a Kantorovich potential for the transport with cost $\frac{1}{2}|x-y|^2$ from $\widehat{\rho}$ to $\rho^\tau_{(k)}$. Actually, the above equality holds $\widehat{\rho}$-a.e., and requires uniqueness of φ. For both these points proving $\rho_\varepsilon > 0$ will be enough, but we will fix this when doing the rigorous proof.

If we combine the fact that the above sum is constant, and that we have $\mathrm{T}(x) = x - \nabla\varphi(x)$ for the optimal T, we get

$$\frac{\mathrm{T}(x) - x}{\tau} = -\frac{\nabla\varphi(x)}{\tau} = \nabla\big(\frac{\delta F}{\delta \rho}(\rho)\big)(x). \tag{8.4}$$

We will denote by $-\mathbf{v}$ the ratio $\frac{\mathrm{T}(x)-x}{\tau}$. Why? Because, as a ratio between a displacement and a time step, it has the meaning of a velocity, but since it is the displacement associated to the transport from $\rho^\tau_{(k+1)}$ to $\rho^\tau_{(k)}$, it is better to view it rather as a backward velocity (which justifies the minus sign).

Since here we have $\mathbf{v} = -\nabla\big(\frac{\delta F}{\delta \rho}(\rho)\big)$, this suggests that at the limit $\tau \to 0$ we will find a solution of

$$\partial_t \rho - \nabla \cdot \big(\rho \nabla\big(\frac{\delta F}{\delta \rho}(\rho)\big)\big) = 0,$$

with homogeneous Neumann boundary conditions on $\partial\Omega$.

Before entering into the details making the above approach rigorous (next section), we want to present some examples of this kind of equations. We will consider two functionals that we already analyzed in Chapter 7, and more precisely

$$\mathscr{F}(\rho) = \int f(\rho(x))\,\mathrm{d}x, \quad \text{and} \quad \mathscr{V}(\rho) = \int V(x)\mathrm{d}\rho.$$

In this case we already saw that we have

$$\frac{\delta\mathscr{F}}{\delta\rho}(\rho) = f'(\rho), \quad \frac{\delta\mathscr{V}}{\delta\rho}(\rho) = V.$$

An interesting example is the case $f(t) = t\log t$. In such a case we have $f'(t) = \log t + 1$ and $\nabla(f'(\rho)) = \frac{\nabla\rho}{\rho}$: this means that the gradient flow equation associated to the functional $\mathscr{F}$ would be the *Heat Equation*

$$\partial_t\rho - \Delta\rho = 0,$$

and that for $\mathscr{F} + \mathscr{V}$ we would have the *Fokker-Planck Equation*

$$\partial_t\rho - \Delta\rho - \nabla\cdot(\rho\nabla V) = 0.$$

We will see a list of other Gradient Flow equations in the Discussion section, including the so-called *Porous Media Equation*, obtained for other choices of f.

Remark 8.4. One could wonder how the usual stochastic diffusion intepretation of the Heat and Fokker-Planck Equations is compatible with the completely deterministic framework that we present here, where the velocity of each particle is given by $-\nabla\rho/\rho - \nabla V$ (take $V = 0$ to get the Heat Equation). Indeed, the Fokker-Planck Equation is also the PDE solved by the density of a bunch of particle following a stochastic equation $dX_t = -\nabla V(X_t)dt + dB_t$, where B is a standard Brownian motion (indepent for each particle). The answer, as it is well pointed out in [89], is based on the idea that in this optimal transport interpretation we rearrange (i.e., relabel) the particles. This fact is quite clear n the 1D case: if at every time step particles move from x to $x - \tau\nabla V(x) + B_\tau$ (i.e. the follow the drift $-\nabla V$ and they diffuse with a Gaussian law), and then we reorder them (we always call particle 1 the one which is the most at the left, particle 2 the next one... and particle N the one which is the most at the right), then we have a discrete family of trajectories which converge, when τ goes to 0 and the number N to infinity, to a solution of Fokker-Planck where the velocity is exactly given by $-\nabla\rho/\rho - \nabla V$. Similar considerations can be done in higher dimension with suitable notions of rearrangement.

Note that all these PDEs come accompanied by Neumann boundary conditions $\rho\frac{\partial}{\partial\mathbf{n}}(\frac{\delta F}{\delta\rho}(\rho)) = 0$ on $\partial\Omega$, as a consequence of the Neumann boundary conditions for the continuity equation of Section 5.3. We will see in Section 8.4.3 a case of extension to Dirichlet boundary conditions.

We finish this section with some philosophical thoughts. Why study some PDEs considering them as gradient flows for the distance W_2? There are at least three reasons.

The first one is that it allows to give an existence result (of a weak solution of such a PDE), with the technique that we will see in the next section. This is obviously useless when the equation is already well-known, as it is the case for the Heat equation, or its Fokker-Planck variant. But the same strategy could be applied to variants of the same equations (or to similar equations in stranger spaces, as it is nowadays done for the Heat flow in metric measure spaces, see [184, 18]). It is also very useful when the equation is of new type (we will see the case of the crowd motion models in Section 8.4.2). The second goal could be to derive properties of the flow and of the solution, once we know that it is a gradient flow. A simple one is the fact that $t \mapsto F(\rho_t)$ must be decreasing in time. However, this can be often deduced from the PDE in different ways and does not require in general to have a gradient flow. The third goal concerns numerics: the discretized scheme to get a gradient flow is itself a sort of algorithm to approximate the solution. If the minimization problems at each time step are discretized and attacked in a suitable way, this can provide efficient numerical tools (see, for instance, [41]).

Finally, all the procedure we presented is related to the study and the existence of weak solutions. What about uniqueness? We stress that in PDE applications the important point is to prove uniqueness for weak solutions of the continuity equation with $\mathbf{v} = -\nabla \frac{\delta F}{\delta \rho}$ (i.e. we do not care at the metric structure, and at the definitions of EVI and EDE). In some cases, this uniqueness could be studied independently of the gradient-flow structure (this is the case for the Heat equation, for instance). Anyway, should we use PDE approaches for weak solutions, or abstract approaches in metric spaces, it turns out that usually uniqueness is linked to some kind of convexity, or λ-convexity, and in particular to displacement convexity. This is why a large part of the theory has been developed for λ-geodesically convex functionals.

8.3 Analysis of the Fokker-Planck case

We consider here the case study of the Fokker-Planck equation, which is the gradient flow of the functional

$$J(\rho) = \int_\Omega \rho \log \rho + \int_\Omega V \mathrm{d}\rho,$$

where V is a Lipschitz function on the compact domain Ω. The initial measure $\rho_0 \in \mathscr{P}(\Omega)$ is taken such that $J(\rho_0) < +\infty$.

We stress that the first term of the functional is defined as

$$\mathscr{F}(\rho) := \begin{cases} \int_\Omega \rho(x) \log \rho(x) \,\mathrm{d}x & \text{if } \rho \ll \mathscr{L}^d, \\ +\infty & \text{otherwise,} \end{cases}$$

where we identify the measure ρ with its density, when it is absolutely continuous. This functional is l.s.c. thanks to Proposition 7.7 (which can be applied since we are on a domain with finite measure, otherwise the situation is trickier, and we refer to **Ex**(45)).

semi-continuity allows to establish the following

Proposition 8.5. *The functional J has a unique minimum over $\mathscr{P}(\Omega)$. In particular J is bounded from below. Moreover, for each $\tau > 0$ the following sequence of optimization*